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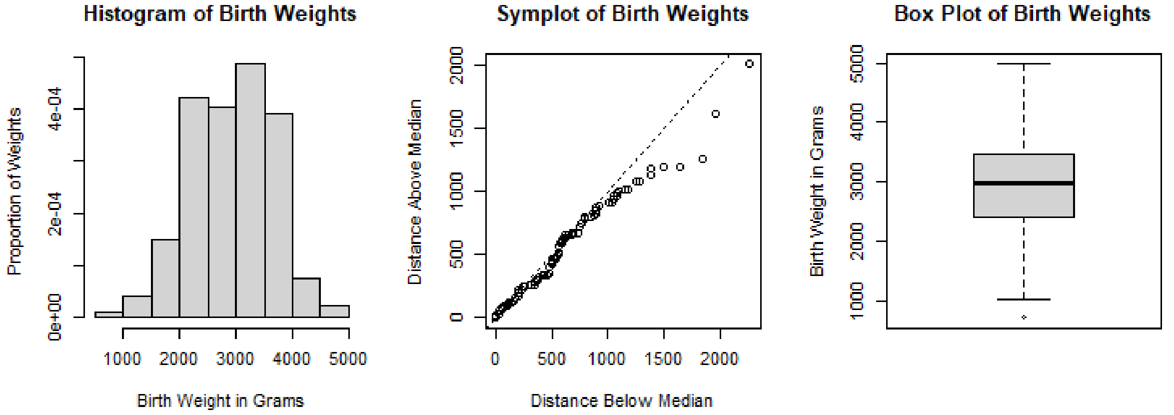
Assignment 4

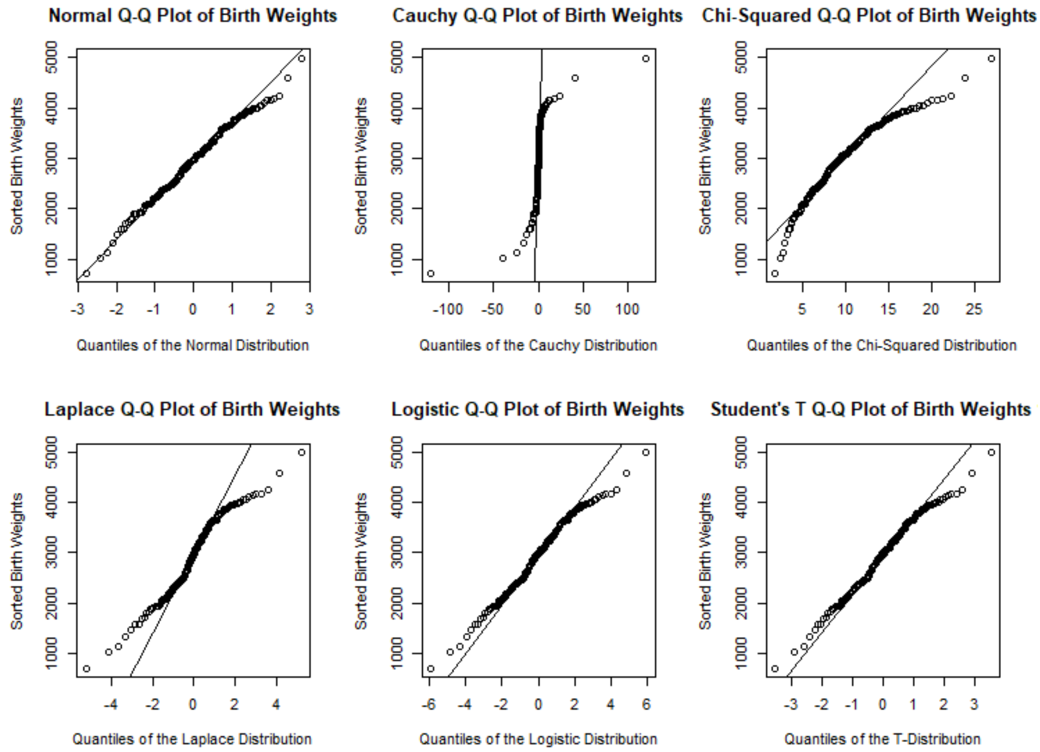
Exercise 4.1

Part a.

The bell-shape of the histogram of birth weights, points on and in the vicinity of the line in the symplot of birth weights, and the symmetry of the box and whiskers above and below the median in the box plot of birth weights suggest that the birth weights are symmetrically distributed.



The distribution of birth weights seems to be in the normal location-scale family because the normal Q-Q plot below is the most linear out of chosen Q-Q plots depicting (quasi-)symmetric distributions and the deviations from the Q-Q line in the normal Q-Q plot are tolerable. Calculating the sample mean and standard deviation suggests that the distribution of birth weights follows $\mathcal{N}(2945, 729^2)$.



Part b.

The function *quantile()* with argument *p*=0.1 is an estimator for the 10% quantile of birth weights. The function gives that the sample 10% quantile is 2038.

The empirical bootstrap is the safest option. In our opinion, however, Part a. of this report substantiates strongly enough that the distribution of birth weight is approximately $\mathcal{N}(2945, 729^2)$. Hence, we opt for the parametric bootstrap. We simulate B=1000 bootstrap samples of size 189 from $\mathcal{N}(2945, 729^2)$. Every bootstrap sample gives a bootstrap value of the estimator 10% quantile. The bootstrap estimate of the median absolute deviation is the median absolute deviation of the 1000 bootstrap values of the estimator 10% quantile. This procedure yields that the bootstrap estimate of the median absolute deviation of the 10% quantile estimator of birth weights is 88.

Part c.

Repeating Part b. with the exponential distribution instead yields that the bootstrap estimate of the median absolute deviation of the 10% quantile estimator of birth weights is 70. Part a. strongly suggests that the distribution of the birth weight is approximately symmetric which means that the assumed exponential distribution is likely incorrect since the exponential distribution is asymmetric. Hence, this wrong parametric distribution leads to big Error 1.

Exercise 4.2

Part a. and b.

B = 2000

	5%	95%
Median	44.5	68.5
Mean	53.296	70,064

Part c.

The biggest difference between the mean is higher that the median and has smaller span.

We prefer the mean method over the median method because of smaller span and the sample mean (61.56) is nice in the middle of the span. This is less the case by the median.

Part d.

	5%	95%
SDRP(mean)	51.522	72,854
CTRP(mean)	59.216	91.939

The interval from these mean tell us that the mean from the CTRP is about 15 higher that the SCRP. Also, the SD is high at the CTRP than the SDRP.

Exercise 4.3

Part a.

See the attachment below Appendix.

Part b. and c.

B = 2000

	1879	1882
P value	0.4896	0.7162
P value test b	0.457	0.6665

The test b is a parametric bootstrap where the D is calculated by ks.test where the input data is a random sample with the same mean and SD as the original sample of the test. The calculation of the p-value percentage of D-values that are smaller than the D-value from the original sample. There is a difference between the two P value for both data sets. But the difference is not a lot. A larger part of the difference is the random samples from the bootstrap.

Appendix

```
# Exercise 4.1
bw = scan("birthweight.txt")

 # Part a.
par(mfrow=c(1,3), pty="s")
hist(bw, prob=T, main="Histogram of Birth Weights",
    xlab="Birth Weight in Grams",
    ylab="Proportion of Weights")
symplot(bw, main="Symplot of Birth Weights")
boxplot(bw, main="Box Plot of Birth Weights",
      ylab="Birth Weight in Grams")

par(mfrow=c(2,3), pty="s")
qqnorm(bw, main="Normal Q-Q Plot of Birth Weights",
      xlab="Quantiles of the Normal Distribution",
      ylab="Sorted Birth Weights")
qqline(bw)

qqcauchy(bw, main="Cauchy Q-Q Plot of Birth Weights",
        xlab="Quantiles of the Cauchy Distribution",
        ylab="Sorted Birth Weights")
qqline(bw, distr=qcauchy)

qqchisq(bw, df=10, main="Chi-Squared Q-Q Plot of Birth Weights",
       xlab="Quantiles of the Chi-Squared Distribution",
       ylab="Sorted Birth Weights")
qqline(bw, distr=function(p) qchisq(p,df=10))

qqlaplace(bw, main="Laplace Q-Q Plot of Birth Weights",
         xlab="Quantiles of the Laplace Distribution",
         ylab="Sorted Birth Weights")
qqline(bw)

qqlogis(bw, main="Logistic Q-Q Plot of Birth Weights",
       xlab="Quantiles of the Logistic Distribution",
       ylab="Sorted Birth Weights")
qqline(bw, distribution=qlogis)

qqt(bw, df=10, main="Student's T Q-Q Plot of Birth Weights",
```

```
   xlab="Quantiles of the T-Distribution",
   ylab="Sorted Birth Weights")
qqline(bw, distr=function(p) qt(p,df=10))

var(bw)
sd(bw)
mean(bw)

  # Part b.
quantile(bw, 0.1)
mad(replicate(1000, quantile(rnorm(length(bw), mean=mean(bw), sd=sd(bw)), 0.1)))

  # Part c.
mad(replicate(1000, quantile(rexp(length(bw), rate=1/mean(bw)), 0.1)))


# Exercize 4.2
  # Part a.
set.seed(2022031839)
B = 2000
median_PRRP = bootstrap(thromboglobulin$PRRP,median,B)
q = c(0.05,0.95)

E42a <- quantile(median_PRRP, q)
E42a
  # Part b.
set.seed(2022031839)
mean_PRRP = bootstrap(thromboglobulin$PRRP,mean,B)

E42b <- quantile(mean_PRRP, q)
E42b
  # Part c.
summary(thromboglobulin$PRRP)
  # Part d.
summary(thromboglobulin$SDRP)
summary(thromboglobulin$CTRP)
set.seed(2022031839)
mean_SDRP = bootstrap(thromboglobulin$SDRP,mean,B)
E42cSDRP <- quantile(mean_SDRP, q)
E42cSDRP
set.seed(2022031839)
mean_CTRP = bootstrap(thromboglobulin$CTRP,mean,B)
E42cCTRP <- quantile(mean_CTRP, q)
E42cCTRP
# Exercize 4.3

  # Part b.
data_1879 <- light[['1879']]
```

```
mean_1879 = mean(data_1879)
length_1879 = length(data_1879)
sd_1879 = sd(data_1879)

D_1879 = ks.test(data_1879, pnorm ,mean_1879, sd_1879)$statistic
D1 = 0.08342437
Tstar = numeric(B)
set.seed(2022031839)
for(i in 1:B){
  xstar = ks.test(rnorm(length_1879, mean_1879, sd_1879), pnorm ,mean_1879, sd_1879)$statistic
  Tstar[i] = xstar
}

c1 = 0
for(i in 1:B){
  if(Tstar[i] > D1){
    c1 = c1 +1
  }
}

p_1879 = c/B
p_1879

data_1882 <- light[['1882']]
mean_1882 = mean(data_1882)
length_1882 = length(data_1882)
sd_1882 = sd(data_1882)
dist_1882 = dist(data_1882)
D_1882 = ks.test(data_1882, pnorm ,mean_1882, sd_1882)$statistic
D_1882
D2 = 0.1453328
Tstar = numeric(B)
set.seed(2022031839)
for(i in 1:B){
  xstar = ks.test(rnorm(length_1882, mean_1882, sd_1882), pnorm ,mean_1882, sd_1882)$statistic
  Tstar[i] = xstar
}

c2 = 0
for(i in 1:B){
  if(Tstar[i] > D2){
    c2 = c2 +1
  }
}

p_1882 = c2/B
p_1879
p_1882
```

```
# Part c.
ks.test(data_1879, pnorm ,mean_1879, sd_1879)
ks.test(data_1882, pnorm ,mean_1882, sd_1882)
```

Exercise 4.3 – Part a.

Suppose the sample $y_1, \dots, y_n$ follows a distribution in the same location-scale family as $X_1, \dots, X_n$. Then $X_i = a + b y_i$ for some $a \in \mathbb{R}$ and $b \in \mathbb{R}^+$. So it must hold that

$$\Phi\left(\frac{X_i - \bar{X}}{s_x}\right) = \Phi\left(\frac{a + b y_i - \frac{1}{n}\sum_{i=1}^{n} a + b y_i}{\sqrt{\frac{1}{n-1}}\sqrt{\sum_{i=1}^{n}\left(a + b y_i - \frac{1}{n}\sum a + b y_i\right)^2}}\right)$$

$$= \Phi\left(\frac{a + b y_i - a - b\bar{y}}{\sqrt{\frac{1}{n-1}}\sqrt{\sum_{i=1}^{n}\left(\cancel{a} + b y_i - \cancel{a} - b\bar{y}\right)^2}}\right)$$

$$= \Phi\left(\frac{\cancel{b} y_i - \cancel{b}\bar{y}}{\sqrt{\frac{1}{n-1}}\sqrt{\sum_{i=1}^{n} \cancel{b^2}(y_i - \bar{y})^2}}\right)$$

$$= \Phi\left(\frac{y_i - \bar{y}}{s_y}\right).$$

That means
$$\tilde{D}_n = \max_i \max\left(\left|\frac{i-1}{n} - \Phi\left(\frac{X_i - \bar{X}}{s_x}\right)\right|, \left|\frac{i}{n} - \Phi\left(\frac{X_i - \bar{X}}{s_x}\right)\right|\right)$$
$$= \max_i \max\left(\left|\frac{i-1}{n} - \Phi\left(\frac{y_i - \bar{y}}{s_y}\right)\right|, \left|\frac{i}{n} - \Phi\left(\frac{y_i - \bar{y}}{s_y}\right)\right|\right).$$

This shows that for any location and scale parameters of the data, the adjusted Kolmogorov-Smirnov test statistic $\tilde{D}_n$ is the same.